Good Things Come in Trees: An Urban Tree's Effect on Crime Mitigation

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In a previous paper estimating the hedonic value of trees, it was found that an additional tree within a postcode in Toronto, roughly equated to a 25 CAD increase in property prices. They found that the heat mitigation effects of trees contributed to around 1 CAD of this value, leaving 24 CAD of unexplained value (Han et al., 2024). The researchers mentioned possible explanations for the discrepancy, listing mental health benefits, pollution reduction, etc., but I hope to quantify more of this discrepancy by answering the question: How much does an urban tree's crime mitigation ability contribute to its total hedonic value?

A common school of thought that many previous studies have supported is that vegetation facilitates crime as it blocks visibility and shields perpetrators from public view. This likely originates from early papers emphasizing the importance of "eyes on the street," and how clear sightlines and pedestrian visibility are key to keeping crime at bay (Jacobs, 1961). This is then emphasized by Oscar Newman's 1972 paper, *Defensible Space: Crime Prevention Through Urban Design,* a well-renowned paper within the urban crime field, which suggests that dense shrubs and poorly maintained landscaping create concealment, and thus higher crime rates. This public sentiment was confirmed as surveys regarding urban parks indicated that citizens felt the least vulnerable in open, mowed areas as opposed to more densely forested ones (Schroeder & Anderson, 1984). When presented with photographs of densely forested parks, citizens also reported that they seemed "dangerous" and "unsafe" (Talbot & Kaplan, 1984). More concretely, one study describes how carjackers utilize vegetation to shield themselves from public view while also using it to discard unwanted goods (Michael, Hull, & Zahm, 1999). From these studies, we can see that the "danger" of vegetation stems from its ability to block visibility and create cover for shady activities.

However, not all vegetation is necessarily vision-blocking. For example, many trees planted in urban areas have higher canopies, and many shrubs are planted low to the ground. Taking this into account, more recent studies have shown a negative relationship between urban vegetation and crime rates in American cities. Specifically, in Chicago, a 10% increase in tree canopy was associated with a 12% decrease in crime, with its effect being most notable in public areas (Troy, Grove, & O'Neil-Dunne, 2012). In Baltimore, researchers found, yet again, an inverse relationship between vegetation and crime, but notably that this relationship experienced diminishing returns. They also noted that this was likely due to two factors: increased surveillance due to people more frequently using outdoor spaces, and preventative treatment due to greenery quelling mental troubles before extreme action takes place (Kuo & Sullivan, 2001). A third paper suggests that trees help reduce crime rates by signaling to criminals that the property is better looked after, and therefore likely has better security measures in place (Donovan & Prestemon, 2012). These implications were also found to have held true in the cities of Philadelphia, Pennsylvania and New Haven, Connecticut, as two studies found that a 10% increase in greenery correlated to a ~15% decrease in crime rates (Wolfe & Mennis, 2012), (Gilstad-Hayden et al., 2015).

Lastly, there has also been significant research regarding crime rate, or rather perceived crime rate, and its effect on property prices. Many researchers have examined property prices in relation to their proximity to sex offenders as listed by Megan's Law, a United States federal law that states that information about sex offenders must be public, including their place of residency. Notably, they found that homes within an 0.1-mile radius from a registered sex offender experience a 2-4% decrease in value (Linden & Rockoff, 2008) (Pope, 2008). Furthermore, one London study claims that when crime rate is perceived higher than the average of neighboring areas, the price of homes in that area experience a value decrease of 10 percent (Gibbons, 2004). This paper was then later reinforced by future researchers as they conducted a similar experiment in the city of Barcelona and found similar numbers of 7-11 percent (Buonanno, Montolio, & Raya-Vilchez, 2012). However, crime rate is not an exogenous independent variable as many of these papers treat it as, and even when it is acknowledged as endogenous, little is done to correct for it. However, one paper utilizes a nine-year crime panel to overcome these errors and found that robbery and aggravated assault crimes were the only category of crimes that influenced property prices, with a 1% increase in violent crimes decrease property prices by 0.25% (Ihlanfeldt & Mayock 2010).

I believe this should be enough to validate the basis of my paper as I've cleared any previous misconceptions and shown that urban forestry affects crime rate, which in turn affects property prices. My proposal's main goal is not to further substantiate this claim, but rather to utilize this claim in quantifying exactly how much crime reduction contributes to a tree's hedonic value. This has yet to be done and will provide further understanding of the priority and necessity of green spaces in urban areas. Obviously, I do not believe that simply planting trees is the most effective route in deterring crime, but I do believe that promoting well-maintained urban greenery has many benefits, two of which being promoting mental wellness while also providing a signal that an area is well-kept, both preemptive factors in preventing crime. As Han's 2024 paper utilizes Toronto's Emerald Ash Borer infestation to isolate exogenous variation in tree canopy, this paper will plan to do the same, but instead looking at its relationship with crime rates over the years and converting that into real value. This conversion can be done through determining the reduction of property prices due to a reduction in crime rate and applying that to a single tree's effect.

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